FM FUNCTOR ON PERFECT COMPLEXES, COMPOSITION, AND ADJOINT KERNELS

THE STACKS PROJECT AUTHORS

Editorial note. Parts (3)-(4) delegate the general adjunction to same-chapter lemmas carrying external citations (uncardable here); the kernel identification, base change, and invertibility arguments are in-span; part (4) is explicitly mirror-level.

Allowed standard prerequisites (editorial).

- FM functor well-definedness. lemma-fourier-mukai (same chapter, judged citation-only, not carded). Part (1) source.
- Composition kernel. lemma-compose-fourier-mukai L26466 (carded this batch). Part (2) source and the K^prime-prime formula.
- General adjunction kernels. lemma-fourier-mukai-right-adjoint, lemma-fourier-mukai-left-adjoint (same chapter, ext-cite filtered, not cardable). Provide the abstract adjoint kernels; identification done in-span.

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FPG: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0C08: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B6S: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A9Y: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 9–9. Navigation pointer

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only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYS: sources/stacks-equiv/pinned-original-source.tex, source lines 1200–1200. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYT: sources/stacks-equiv/pinned-original-source.tex, source lines 1282–1282. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYX: sources/stacks-equiv/pinned-original-source.tex, source lines 1577–1577. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYW: sources/stacks-equiv/pinned-original-source.tex, source lines 1487–1487. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B6F: focused-seeds/geometry/S9-eqv-fourier-mukai-flat-proper-over-noetherian/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a Noetherian scheme.*

1. *For X, Y proper and flat over S and K in $D_{perf}(\mathcal{O}_{X\times_S Y})$ we obtain a Fourier-Mukai functor $\Phi_K : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$.*
2. *For X, Y, Z proper and flat over S, $K \in D_{perf}(\mathcal{O}_{X\times_S Y})$, $K' \in D_{perf}(\mathcal{O}_{Y\times_S Z})$ the composition $\Phi_{K'} \circ \Phi_K : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Z)$ is equal to $\Phi_{K''}$ with $K'' \in D_{perf}(\mathcal{O}_{X\times_S Z})$ computed as in Lemma 0FYS,*
3. *For X, Y, K, Φ_K as in (1) if $X \to S$ is Gorenstein, then $\Phi_{K'} : D_{perf}(\mathcal{O}_Y) \to D_{perf}(\mathcal{O}_X)$ is a right adjoint to Φ_K where $K' \in D_{perf}(\mathcal{O}_{Y\times_S X})$ is the pullback of $L\mathrm{pr}_1^*\omega_{X/S}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K^\vee$ by $Y \times_S X \to X \times_S Y$.*
4. *For X, Y, K, Φ_K as in (1) if $Y \to S$ is Gorenstein, then $\Phi_{K''} : D_{perf}(\mathcal{O}_Y) \to D_{perf}(\mathcal{O}_X)$ is a left adjoint to Φ_K where $K'' \in D_{perf}(\mathcal{O}_{Y\times_S X})$ is the pullback of $L\mathrm{pr}_2^*\omega_{Y/S}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K^\vee$ by $Y \times_S X \to X \times_S Y$.*

Proof. Part (1) is immediate from Lemma 0FYT part (4).

Part (2) follows from Lemma 0FYS and the fact that $K'' = R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y\times_S Z}} L\mathrm{pr}_{23}^*K')$ is perfect for example by Derived Categories of Schemes, Lemma 0B6F.

The adjointness in part (3) on all complexes with quasi-coherent cohomology sheaves follows from Lemma 0FYW with K' equal to the pullback of $R\mathcal{H}om_{\mathcal{O}_{X\times_S Y}}(K, a(\mathcal{O}_Y))$ by $Y \times_S X \to X \times_S Y$ where a is the right adjoint to $R\mathrm{pr}_{2,*} : D_{QCoh}(\mathcal{O}_{X\times_S Y}) \to D_{QCoh}(\mathcal{O}_Y)$. Denote $f : X \to S$ the structure morphism of X. Since f is proper the functor $f^! : D^+_{QCoh}(\mathcal{O}_S) \to D^+_{QCoh}(\mathcal{O}_X)$ is the restriction to $D^+_{QCoh}(\mathcal{O}_S)$ of the right adjoint to $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_S)$, see Duality for Schemes, Section 0A9Y. Hence the relative dualizing complex $\omega^\bullet_{X/S}$ as defined in Duality for Schemes, Remark 0B6S is equal to $\omega^\bullet_{X/S} = f^!\mathcal{O}_S$. Since formation of the relative dualizing complex commutes with base change (see Duality for Schemes, Remark 0B6S) we see that $a(\mathcal{O}_Y) = L\mathrm{pr}_1^*\omega^\bullet_{X/S}$. Thus

$$R\mathcal{H}om_{\mathcal{O}_{X\times_S Y}}(K, a(\mathcal{O}_Y)) \cong L\mathrm{pr}_1^*\omega^\bullet_{X/S} \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K^\vee$$

by Cohomology, Lemma 08DQ. Finally, since $X \to S$ is assumed Gorenstein the relative dualizing complex is invertible: this follows from Duality for Schemes, Lemma 0C08. We conclude that $\omega^\bullet_{X/S}$ is perfect (Cohomology, Lemma 0FPG) and hence K' is perfect. Therefore $\Phi_{K'}$ does indeed map $D_{perf}(\mathcal{O}_Y)$ into $D_{perf}(\mathcal{O}_X)$ which finishes the proof of (3).

The proof of (4) is the same as the proof of (3) except one uses Lemma 0FYX instead of Lemma 0FYW. □